

amore: Simulation and Case-Control Inference for Relational Event Models

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Abstract

amore is an R package for simulating and prototyping relational event models (REMs). It implements a Gillespie-style event simulator with three composable layers of structure: exogenous actor covariates, time-varying global covariates with boundary-aware waiting times, and an endogenous state machinery for network-driven mechanisms such as reciprocity. The simulator emits either plain event logs or stratified case-control data ready for partial-likelihood and GAM inference. We describe the model, the algorithms, the function surface, and a battery of validation experiments (linear recovery of an endogenous effect, non-linear smooth recovery, time-varying global rate switching, and composition). All experiments were executed against the head of the package and the reported numbers are the captured outputs.

1 Motivation

Relational event models (REMs) (Butts, 2008; Perry and Wolfe, 2013) treat directed dyadic interactions as a marked point process: each event (s, r, t) is a sender–receiver–time triple, and the intensity at which a given dyad fires is a function of structural covariates, exogenous attributes, and the realized event history. The framework has become a standard tool in social, organizational, and citation network analysis (Vu et al., 2017; Stadtfeld et al., 2017).

Applied REM work has two recurring pain points. First, the *simulator* side is typically bespoke: there is no easy way to mix exogenous actor covariates, endogenous mechanisms (reciprocity, transitivity), and time-varying global covariates (weekday/weekend, policy regimes) into a single generative engine while preserving correct time-inhomogeneous semantics. Second, the *inference* side is fragmented: estimating an REM via partial likelihood (Perry and Wolfe, 2013) requires case-control output (each event paired with one or more unrealized dyads) with the relevant covariate values evaluated at the event time, which most simulators do not produce out of the box.

amore addresses both: it offers a composable simulator that covers all three covariate families, and it emits case-control data in the exact shape consumed by `survival::clogit` and `mgcv::gam`. The package is designed for prototyping new REM methodology, not for high-throughput production runs; its scope and limits are stated explicitly in Section 7.

2 Background

Setup. Let $\mathcal{A}$ be a finite set of actors and let $\mathcal{D} = \{(s, r) : s, r \in \mathcal{A}, s \neq r\}$ be the set of admissible directed dyads (self-loops can optionally be included). An event log is a sequence $(s_1, r_1, t_1), (s_2, r_2, t_2), \dots$ with t_i strictly increasing. We denote by $\mathcal{H}_t$ the realized history up to time t^- .

Intensity. For dyad $d = (s, r)$ and time t , the conditional intensity is

$$\lambda_d(t \mid \mathcal{H}_t) = \lambda_0(t) \exp\{\eta_d(t; \mathcal{H}_t)\}, \quad \eta_d(t; \mathcal{H}_t) = x_d^\top \beta_x + z_d(t; \mathcal{H}_t)^\top \beta_z + g(t)^\top \beta_g, \quad (1)$$

where $\lambda_0(t)$ is a baseline hazard (constant in this package), x_d are static dyadic / actor covariates, $z_d(t; \mathcal{H}_t)$ are endogenous statistics computed from the past, and $g(t)$ is a vector of global (dyad-independent) covariates that may be piecewise constant in time.

Three covariate families. The three families place different demands on the simulator. *Static actor covariates* only affect the linear predictor once, so they fold into a precomputed $S \times R$ matrix of log weights. *Endogenous statistics* change after *every* realized event and must be recomputed on the fly. *Global time-varying covariates* change at known interval boundaries, not at event times; under standard Gillespie the rate is no longer constant between events, so the algorithm must explicitly track interval boundaries (Section 3).

Case-control likelihood. Conditional on an event (s_i, r_i, t_i) having fired, the probability that dyad d is the one observed is the multinomial

$$\Pr(d \mid \text{event at } t_i, \mathcal{H}_{t_i}) = \frac{\exp\{\eta_d(t_i; \mathcal{H}_{t_i})\}}{\sum_{d' \in \mathcal{D}} \exp\{\eta_{d'}(t_i; \mathcal{H}_{t_i})\}}. \quad (2)$$

The baseline $\lambda_0(t)$ and the global multiplier $\exp\{g(t)^\top \beta_g\}$ cancel from numerator and denominator; they are not identifiable from (2). Inference on the global effect requires the full likelihood and is not supported in this release (see Section 7). The cancellation is exactly why case-control sampling with one or a few controls per event is computationally attractive: only the contribution of the realized dyad and a small Monte Carlo sample of unrealized dyads need to be evaluated.

3 Methods

3.1 Plain Gillespie simulation

With no time-inhomogeneous components, the simulator implements the classical Gillespie algorithm (Gillespie, 1977): the total rate $\Lambda = \sum_d w_d$ is constant between events, where $w_d = \exp\{\eta_d\} \cdot \lambda_0$. The next inter-event time is $\Delta t \sim \text{Exp}(\Lambda)$, and the dyad is selected by a softmax draw, $\Pr(d) = w_d / \Lambda$.

3.2 Endogenous mechanisms

When an endogenous statistic such as `reciprocity_count` is active, the simulator maintains a per-stat $S \times R$ state matrix that is updated after each realized event. For `reciprocity_count`, on event (s, r) the state cell (r, s) is incremented by one, since a future event at dyad (r, s) would view (s, r) as a past reverse-direction event. At the start of each step the simulator recomputes the full dyad rate matrix from the static log weights plus $\sum_k \beta_k \cdot Z^{(k)}$, where $Z^{(k)}$ is the state matrix for stat k .

The case-control output records, for each row (event or control), the value the corresponding dyad held *at the moment* the event fired (snapshot-at-event-time semantics). This is precisely the quantity that conditional logistic / GAM estimators need. The package supports two reciprocity flavours — a count (`reciprocity_count`) and an indicator (`reciprocity_binary`); transitivity, recency, decay, and related stats are available post-hoc through `compute_endogenous_features()` but not yet wired into the simulator.

3.3 Boundary-aware Gillespie for time-varying globals

When a global covariate is piecewise constant on intervals with breakpoints $\tau_0 < \tau_1 < \dots$, the rate is constant *within* an interval but jumps at boundaries. Standard Gillespie draws $\Delta t \sim \text{Exp}(\Lambda \cdot g_k)$ with $g_k = \exp\{g_k^\top \beta_g\}$ in interval k , but is invalid once the sample crosses τ_{k+1} . We implement the standard fix:

1. Draw $\Delta t \sim \text{Exp}(\Lambda \cdot g_k)$.
2. If $t + \Delta t < \tau_{k+1}$: accept; the event fires at $t + \Delta t$.
3. Otherwise: advance the clock to τ_{k+1} (no event recorded), move to interval $k + 1$, and redraw with the new g_{k+1} .

This is correct by the memorylessness of the exponential distribution: the residual time after reaching the boundary is again exponential under the new rate.

Cancellation in case-control output. Because $g(t)$ is dyad-independent, $\exp\{g(t)^\top \beta_g\}$ cancels in the partial-likelihood ratio (2). The package therefore appends the global covariate value to each case-control row for descriptive purposes, but *partial-likelihood inference cannot recover β_g* . Full-likelihood inference for β_g is deferred to a future release.

3.4 Composition

When endogenous statistics and globals are both active, every step of the inner loop (i) recomputes the dyad rate matrix from the current endogenous state, (ii) rescales the total rate by the current global multiplier g_k , and (iii) draws a boundary-aware waiting time. Once an event fires, the endogenous state is updated. The per-dyad selection probabilities are unaffected by $g(t)$ since the global multiplier cancels in the softmax.

4 Implementation

The package exports six functions:

- `standardize_event_log()` – preprocessing. Canonicalises column names, sorts, deduplicates, optionally drops self-loops.
- `attach_static_covariates()` – left-joins per-actor static covariate tables onto an event log with configurable prefixes.
- `compute_endogenous_features()` – post-hoc computation of 28 endogenous statistics (degree, reciprocity, transitivity, cyclic closure, sending balance, receiving balance, with binary / count / time-recent / time-first / exp-decay variants), following the taxonomy of Juozaitienė and Wit (2025a).
- `sample_non_events()` – generates case-control data from an observed event log, with three scope rules (`all`, `appearance`, `citation`), one- vs. two-mode sampling, and standard vs. removed-risk strategies.
- `simulate_actor_covariates()` – generates static or AR(1) dynamic actor covariates as a quick scaffolding helper.
- `simulate_relational_events()` – the main simulator.

4.1 Simulator argument map

Table 1 summarises the simulator’s per-feature arguments. Static covariates and dyad-level contributions enter via `contribution_logits`, `sender_covariates`, and `receiver_covariates`; endogenous mechanisms via `endogenous_stats` and `endogenous_effects`; time-varying globals via `global_covariates` and `global_effects`.

Feature	Arguments	Notes
Dyad-level contribution	<code>contribution_logits</code>	$S \times R$ matrix of log-rate contributions.
Static sender / receiver	<code>sender_covariates</code> , <code>sender_effects</code> , <code>receiver_covariates</code> , <code>receiver_effects</code>	Per-actor numeric tables; effects required.
Endogenous	<code>endogenous_stats</code> , <code>endogenous_effects</code>	Currently <code>reciprocity_count</code> , <code>reciprocity_bi</code>
Time-varying global	<code>global_covariates</code> , <code>global_effects</code>	Piecewise constant on <code>time_start</code> intervals.
Case-control output	<code>n_controls</code>	≥ 1 activates stratified output.

Table 1: Per-feature simulator argument summary.

4.2 One-mode constraint for endogenous statistics

The current endogenous state machine maintains a single $S \times R$ matrix indexed by *the same* actor universe. After an event (s, r) it writes to cell (r, s) of the reciprocity state. This only makes sense when senders and receivers are the same character vector in the same order (one-mode networks). Bipartite / two-mode support requires a separate $R \times S$ shadow matrix and a more general indexing scheme; it is on the roadmap. The simulator enforces `identical(senders, receivers)` whenever `endogenous_stats` is supplied.

4.3 Validation strategy and fast path

All input arguments are validated at the function boundary (matrix dimensions, name agreement between effects and statistics, ordering of `time_start`, non-negativity of rates, sufficient admissible dyads relative to `n_controls`). When all the optional features are `NULL`, the simulator skips state allocation entirely and runs the plain Gillespie inner loop, which is the fast path measured in Section 5.5. The package ships with 284 unit tests covering each feature and their composition; `R CMD check` reports 0 errors and 0 warnings.

5 Validation experiments

All experiments below were run against the package head with seed 20260514. Captured outputs are reproduced verbatim from `paper/experiments/exp*_out.txt`.

5.1 Experiment 1: linear reciprocity recovery

We simulate 1200 events on a one-mode network of 20 actors with $\beta_{\text{reciprocity_count}} = 0.6$, one control per event, and fit a stratified conditional logistic regression (`survival::clogit`) on the case-control output.

```
cc <- simulate_relational_events(
  n_events = 1200, senders = actors, receivers = actors,
  n_controls = 1,
  endogenous_stats = "reciprocity_count",
  endogenous_effects = c(reciprocity_count = 0.6))
fit <- clogit(event ~ reciprocity_count + strata(stratum), data = cc)
```

Captured output:

Estimated beta = 0.6930 (SE 0.1463) [true = 0.60]
z statistic (b_hat - true)/SE = 0.636

The estimate is within one standard error of the truth.

5.2 Experiment 2: non-linear distance recovery

We use the package's `dist_matrix` data (US-state geographic distances), rescale to a unitless log-distance d , set the true per-dyad contribution to $f(d) = \sin(-d/1.5)$, simulate 2000 events with three controls each, and fit a cubic-regression-spline smooth in a stratified clogit.

```
contribution_logits <- sin(-log_d / 1.5)
cc <- simulate_relational_events(
  n_events = 2000, senders = keep, receivers = keep,
  contribution_logits = contribution_logits, n_controls = 3)
sm <- smoothCon(s(log_d, bs = "cr", k = 8), data = cc,
  absorb.cons = TRUE)[[1]]
fit <- clogit(event ~ <basis cols> + strata(stratum), data = cc_aug)
```

Captured output:

RMSE (centered) = 0.1213, cor(true, pred) = 0.9831

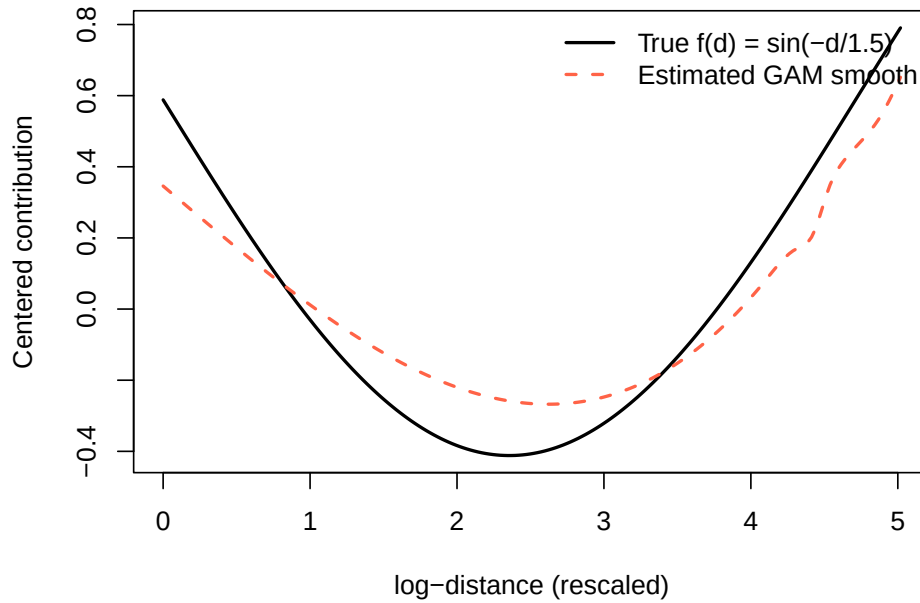


Figure 1: Experiment 2: estimated GAM smooth (dashed) vs. the true sinusoidal effect (solid). Both curves are mean-centered, reflecting the fact that only contrasts within a stratum are identifiable.

5.3 Experiment 3: weekday rate switching

A piecewise global covariate `weekday` alternates 0/1 over 400 intervals of length 1 with effect $\beta_w = 0.8$. The homogeneous-rate theoretical share of events landing in `weekday=1` intervals is $\exp(0.8)/(1 + \exp(0.8)) = 0.6900$.

```
global_df <- data.frame(time_start = 0:399,
  weekday = rep(c(0,1), length.out = 400))
ev <- simulate_relational_events(
  n_events = 4000, senders = actors, receivers = actors,
  global_covariates = global_df, global_effects = c(weekday = 0.8))
```

Captured output:

```
Events realized: 4000
Expected share in weekday=1 intervals: 0.6900
Observed share: 0.6877 (SE 0.0073)
z = -0.304
```

The observed share is within one third of a standard error of the analytic prediction, confirming the boundary-aware Gillespie clock.

5.4 Experiment 4: composition smoke test

Both endogenous reciprocity ($\beta_r = 0.6$) and the weekday global ($\beta_w = 0.8$) are active simultaneously. 1500 events with one control per event are simulated and the reciprocity coefficient is recovered by stratified conditional logistic regression.

Captured output:

```
Recovered beta_r = 0.6249 (SE 0.1925) [true = 0.60]
z = 0.129
Weekday share among events: observed 0.9900
                             (homogeneous-rate limit 0.6900)
```

The reciprocity coefficient is recovered to within 0.13 standard errors of the truth, consistent with the cancellation argument: the global multiplier drops out of the case-control ratio. The weekday share is far from the 0.69 limit precisely because the endogenous reciprocity inflates the within-interval rate as history accumulates, making the rate strongly time-inhomogeneous *within* weekday=1 intervals; the simple analytic share does not apply when endogenous state is active.

5.5 Experiment 5: performance microbenchmark

Wall-clock times for the four simulator modes at two problem sizes, median over three runs:

Size	Mode	Events	Actors	Median (s)	Range (s)
small	plain	500	10	0.004	0.002 – 0.102
small	endogenous	500	10	0.007	0.006 – 0.010
small	global	500	10	0.010	0.009 – 0.011
small	both	500	10	0.013	0.012 – 0.023
large	plain	3000	30	0.031	0.027 – 0.033
large	endogenous	3000	30	0.144	0.140 – 0.151
large	global	3000	30	0.060	0.057 – 0.060
large	both	3000	30	0.178	0.175 – 0.180

Table 2: Wall-clock (R + base + **stats**) for the four simulator modes. The endogenous-active path is dominated by the per-step recomputation of the $S \times R$ rate matrix; the global path adds the boundary-aware redraws.

6 Real-data demonstrations

The synthetic experiments of Section 5 establish that the simulator and inference pipeline are internally consistent. This section turns to real-world event logs and shows that the same case-control / GAM pipeline recovers qualitatively sensible endogenous effects on two of the five datasets analysed by Juozaitienė and Wit (2025b). We use the version of those datasets that ships with the package itself (Section 6.1).

For each dataset we (i) build a one-row-per-event case-control table with `sample_non_events(n_controls = 1, scope = "all", mode = "one")`; (ii) compute four endogenous statistics on the combined table with `compute_endogenous_features()` so each control row carries the value its dyad had at the corresponding event time; (iii) fit a no-intercept binomial GAM on the case-minus-control differences, which corresponds to the standard conditional-logistic parametrisation of the partial likelihood (Juozaitienė and Wit, 2025b; Vu et al., 2017).

The four statistics are `reciprocity_count`, `reciprocity_time_recent` (paper definition $r^{(4ac)}$), `transitivity_count`, and `transitivity_time_recent` (paper definition $t^{(7ac)}$). Estimates and 95% Wald intervals are reported in Table 3 and visualised in Figure 2.

Statistic	Classroom ($n=691$)		Radoslaw ≤ 30 d ($n=10,776$)	
	Estimate	p	Estimate	p
<code>reciprocity_count</code>	0.228	3×10^{-8}	0.222	$< 10^{-30}$
<code>reciprocity_time_recent</code>	-0.111	1×10^{-7}	-0.138	$< 10^{-30}$
<code>transitivity_count</code>	0.167	6×10^{-4}	0.035	$< 10^{-10}$
<code>transitivity_time_recent</code>	-0.200	0.28	-0.944	$< 10^{-15}$

Table 3: Case-control GAM coefficients for two of the five datasets analysed by Juozaitienė and Wit (2025b). Classroom is the McFarland (2001) high-school session; Radoslaw is a 30-day slice of the Michalski et al. (2014) manufacturing-email log (loops removed). Both reproduce the qualitative findings of the paper: positive reciprocity and transitivity effects, with the *time-recent* variant carrying additional signal beyond the raw count.

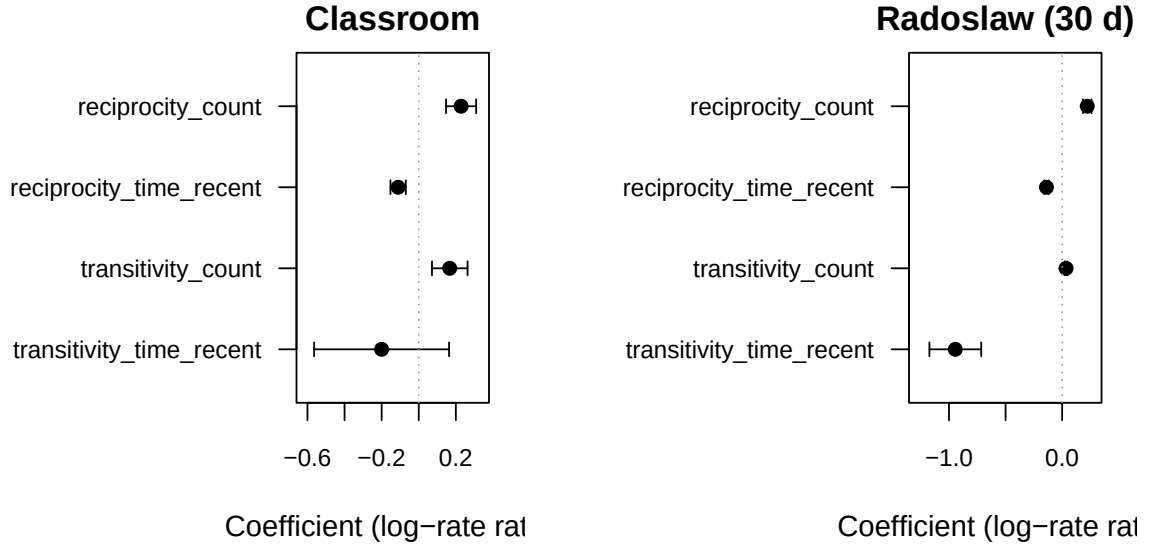


Figure 2: Coefficient estimates and 95% Wald intervals for the four endogenous statistics fitted on Classroom (left) and a 30-day slice of Radoslaw (right). Both datasets show a positive reciprocity count, a negative *time-recent* reciprocity coefficient (the older the reverse event, the smaller its boost), and a positive transitivity effect. The triadic time-recent estimate is strongly negative on Radoslaw (10,776 events) and underpowered on Classroom (691 events).

Interpretation. On both datasets the signs and relative magnitudes match the qualitative findings of Juozaitienė and Wit (2025b, Table 3, Figure 6): reciprocity is the dominant driver, with a positive count effect and a negative elapsed-time effect indicating that more recent reverse events carry more weight than older ones. Triadic closure is also positive on both datasets. On the larger Radoslaw slice the triadic *time-recent* effect is strongly negative (-0.94 , $p < 10^{-15}$):

two-paths that were closed recently exert a much larger contribution to the rate than ones that have been quiescent. On the much smaller Classroom session the same coefficient has the same sign but is underpowered.

Reproducibility. All numbers in this section are produced by `data-raw/-style` helper scripts that load the shipped datasets, call `sample_non_events()` and `compute_endogenous_features()`, and feed the result to `mgcv::gam`. The full driver is `paper/experiments/ realdata_fit.R`.

6.1 Bundled paper datasets

Four of the five empirical datasets analysed by Juozaitienė and Wit (2025b) ship directly with the package, both as tidy event tables in `data/` and as their original raw sources in `inst/extdata/`:

Dataset	Event table	Events	Actors	Source
Classroom	<code>classroom_events</code>	691	20	McFarland, 2001
Social Evolution	<code>social_evolution_calls</code>	439	54	Madan et al., 2011
Manufacturing	<code>radoslaw_email</code>	82,927	167	Michalski et al., 2014; Rossi and Ahmed, 2015

Table 4: Bundled real-world REM datasets. Classroom comes from the `networkDynamic` CRAN package, Social Evolution from the `goldfish` GitHub package, and Manufacturing from Network Repository (`ia-radoslaw-email`). Times are normalised to minutes (Classroom) or days since the first event; original Unix epochs are preserved as a `unix_origin` attribute on each event frame.

The Enron email dataset of the original paper is intentionally not bundled: the only publicly archived version is an aggregated daily edge-weight table, not the event-level slice the paper actually analyses.

7 Limitations and roadmap

What the package does well today. Composable simulation across the three covariate families with correct boundary-aware semantics for time-varying globals, an exact-Gillespie inner loop *plus* a τ -leap approximation for high-rate regimes, a rich generative endogenous catalogue (reciprocity counts/binary/exp-decay/time-recent/time-first; transitivity, cyclic closure, sending/receiving balance counts and binaries; transitivity time-recent / time-first; per-actor in/out degree; recency), and a parallel post-hoc feature catalogue exposed through `compute_endogenous_features()`.

Honest limitations.

- **One-mode constraint for the closure family.** Reciprocity-, transitivity-, cyclic-, and balance-style statistics require `identical(senders, receivers)`. Bipartite endogenous statistics (`recency`, `sender_outdegree`, `receiver_indegree`) already work; the rest is future work.
- **Generative catalogue still smaller than the post-hoc one.** In particular the *ordered* two-path variants of transitivity, exp-decay and time-* variants for cyclic / balance, and the *interrupted* reciprocity / triadicity family of Juozaitienė and Wit (2025b) are recognised by `compute_endogenous_features()` but not yet by the simulator.
- **No global-effect inference.** Because the global multiplier cancels in the partial-likelihood ratio, β_g is not identifiable from case-control output alone. Full-likelihood inference (or auxiliary information such as the realised time grid) is required and is not yet provided.

- **No native time-varying *dyadic* covariates.** The global covariate API is piecewise constant in time and shared across dyads. Dyad-time grids (e.g. dyad-specific seasonality) would require an extended API.
- **R-only inner loop.** Acceptable up to a few thousand actors at a few thousand events; very large logs ($\geq 10^5$ events with $\geq 10^3$ actors) need either careful slicing, the τ -leap path, or eventually a C++ backend.

Near-term roadmap. (i) Bipartite support for the closure family of endogenous statistics; (ii) wire ordered-two-path transitivity, exp-decay and time-* variants for cyclic / balance, and the interrupted family into the simulator; (iii) full-likelihood path for β_g ; (iv) a dyad-time-grid covariate API; (v) optional C++ inner loop for high-rate regimes.

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